

FIT3155 SAMPLE EXAM PAPER

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INSTRUCTIONS

- This exam is worth 60 marks.

General exam technique

- Some students lose marks by not attempting **all** questions.
- Suppose you are likely to get 3/6 on a question after spending say 10 minutes on it. Spending another 10 minutes on the same question may gain you *at most* 3 additional marks. In contrast, spending those 10 minutes on a new question could earn you another 6 marks. The key point is: do not get stuck on one question.
- A good strategy is to first answer the questions you are most confident about before attempting the others. In other words, answering questions strictly serially during an exam can be suboptimal.
- Do not waste time answering an unrelated question in the hope of gaining partial marks. Marking will be based on the question that was actually asked.
- Where necessary, justify your answers with clear and concise explanations, without being overly wordy.

Question 1 (6 marks)

A string $S[1 \dots n]$ has a **full period** of length p if and only if $S[1 \dots p]$ is the **shortest** prefix of S such that S can be obtained by repeating this prefix k times, i.e., $k \times p = n$, where n , p , and k are positive integers. In particular, if no shorter prefix repeats to form S , the full period is $p = n$, corresponding to the string repeating once.

Example: $S[1 \dots 24] = \text{abcdabcdabcdabcdabcdabcd}$ has a period length of $p = 4$, since the prefix **abcd** repeats $k = 6$ times.

Write pseudocode describing an **efficient** algorithm that accepts any input string $S[1 \dots n]$ and outputs its full period.

(For this problem, you may assume that `zalg` is available as a built-in function that takes a string as input and returns the array of its Z -values. Do not write pseudocode for `zalg`.)

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Question 2

(6 marks)

In the Boyer–Moore algorithm for exact matching of the pattern $\text{pat}[1 \dots m]$ against the text $\text{txt}[1 \dots n]$, consider a general iteration in which the region $\text{txt}[j \dots j + m - 1]$ of the text is being compared with pat , and a right-to-left scan results in a mismatch at position $1 \leq k \leq m$, where the value in the goodsuffix (gs) array satisfies $\text{gs}[k + 1] > 0$.

Prove that the good suffix shift of $m - \text{gs}[k + 1]$ positions to the right is **safe**, i.e., such a shift can never skip pat past any valid occurrence in txt .

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Question 3

(6 marks)

Let $S[1 \dots n-1]\$$ be a string whose Burrows–Wheeler Transform (BWT) is $BWT[1 \dots n]$. (Assume that the alphabet of S is Σ , and that $\$$ is a unique terminal symbol.)

Write pseudocode to construct a data structure that, for each $1 \leq i \leq n$, stores the number of times each character $BWT[i]$ has been observed in the range $BWT[1 \dots i-1]$ (both endpoints inclusive).

Characterize the worst-case time and space complexity of the construction.

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These questions relate to Ukkonen's linear time suffix tree construction algorithm of any given string $S[1...n]$.

- (i) Can a new internal node be created (using Rule 2 regular) in the existing implicit suffix tree during the last suffix extension ($j = i$) of any general phase i , where $2 \leq i \leq n$? Answer Yes or No and then justify your answer.
- (ii) If Rule 3 was applied to complete the suffix extension j in phase i of the algorithm. For the same extension j in the very **next phase** $i + 1$, would it be possible to potentially apply Rule 1 ('leaf-edge extension rule')? Answer Yes or No and then justify your answer.

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Question 5

(6 marks)

1. Work out the Elias-Omega binary codeword for positive integer 15.
2. The following tuples were derived using the LZ77 method of encoding:

[0,0,p]

[0,0,q]

[0,0,r]

[3,3,s]

[7,6,t]

Decode these LZ77 tuples to regenerate the original string.

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Question 6

(6 marks)

Describe the steps that Miller-Rabin's randomized method uses to test primality of any given integer n .

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Question 7

(6 marks)

Write a pseudocode describing the **merge** function that takes two Fibonacci heaps as inputs and returns a merged Fibonacci heap as output.

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Question 8

(6 marks)

A k -bit binary counter data structure supports an **increment** by 1 operation (modulo 2^k), starting with all k bits of the counter initialized to 0.

Using the aggregate method of amortized analysis, determine the amortized time complexity of an **increment** operation for this data structure.

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Question 9

(6 marks)

For any B-tree with minimum degree t that contains N keys, derive an upper bound on the worst-case height of the B-tree (in terms of N and t).

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Question 10

(6 marks)

Solve the following optimization problem using the simplex algorithm.

$$\text{Maximize } z = x + y$$

Subject to the constraints:

$$x + 2y \leq 4$$

$$4x + 2y \leq 12$$

$$-x + y \leq 1$$

$$x \geq 0, y \geq 0$$

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